

Adrian Gonçalves Lopes

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OBJECTIVE

Student seeking my first professional opportunity in the technology field, focusing on Junior Backend Developer, Junior Full-Stack, or Technical Support positions. Great teamwork skills, easily adapt to remote work (strong commitment to schedule and meetings), and fast learner. Highly motivated to apply my knowledge, collaborate with talented people, and continue growing professionally.

EDUCATION

Centro Estadual Senador Jesse Pinto Freire — Technical High School in Information Technology

Period: 2021 – 2023

UNINASSAU (In progress) — Bachelor's Degree in Systems Analysis and Development (ADS)

Period: 2026 – 2027

TECHNICAL SKILLS

Languages: Python · Java · JavaScript · HTML · CSS · SQL

Frameworks & Libraries: Spring Boot · Spring AI · Spring Data JPA · Next.js 15 · Node.js · Streamlit · Thymeleaf

Machine Learning & Data: Pandas · Ensemble models (Random Forest, Logistic Regression, Gradient Boosting) · Model explainability with SHAP values

Tools: Maven · pdfplumber · RabbitMQ · GitHub Actions (CI/CD) · SpringDoc OpenAPI / Swagger UI · python-dotenv · streamlit-authenticator

Databases: PostgreSQL · pgvector (vector extension for RAG/embeddings) · H2 Database

Others: OpenAI API (GPT-4o-mini) · RAG (Retrieval-Augmented Generation) · Microservices Architecture · Event-Driven Messaging (Topic Exchange) · JWT/bcrypt (authentication) · Automated Testing (JUnit 5, Mockito, MockMvc) · Regex PDF parsing · WhatsApp Web integration (CRM) · REST API · Multi-stage Dockerfile

PERSONAL PROJECTS / PORTFOLIO

CARTEC 2.0

- ERP/CRM platform in production for automotive workshops, running on a self-managed Linux VPS. Integrates AI agents via WhatsApp using n8n and Evolution API, Claude Vision API for document/image analysis, JWT/bcrypt security, and nginx reverse proxy.

MatchMind AI (Now or Never)

- Football predictive intelligence platform, with probability generation calibrated for 9 different markets per match and explainability via SHAP values.
- Backend in Spring Boot (Java 17+) controlling persistence (H2) and orchestrating the Machine Learning pipeline via asynchronous invocation.
- ML pipeline in Python using ensemble models (Random Forest, Logistic Regression, Gradient Boosting), dynamic feature engineering from official standings, and text-based explanations generation (PT-BR) based on SHAP impacts; includes smart rate limiting (10 req/min) with auto-retry.
- Frontend with dynamic dashboard in Next.js 15 displaying probabilities and analytical charts.

Link: <https://github.com/AdrianLopez001/Now-or-Never>

Corporate RAG Engine

- Enterprise-grade RAG (Retrieval-Augmented Generation) backend, built with the Java AI stack (Java 21, Spring Boot 3.3, Spring AI, GPT-4o-mini, PostgreSQL + pgvector with HNSW indexing, Docker, Apache Tika).
- CI pipeline configured with GitHub Actions and automated testing with Mockito and mocked context loading.

Link: <https://github.com/AdrianLopez001/Corporate-RAG-Engine>

CareSync

- Event-driven appointment scheduling and notification system, built with independent microservices (appointment-service and notification-service) communicating asynchronously via RabbitMQ (Topic Exchange).
- PostgreSQL database with transactional consistency ensuring post-commit publication; unit testing (JUnit 5/Mockito) and web API testing (MockMvc) integrated in a CI pipeline (GitHub Actions).

Link: <https://github.com/AdrianLopez001/CareSync>

System Lumiar

- Modern charity platform for volunteers and social campaign management for non-profit organizations, developed using Next.js and TypeScript.

Link: <https://github.com/AdrianLopez001/System-lumiar>

Fintrack

- Python backend system for personal/business financial control and audit, focusing on financial data integrity, bank statement processing, and smart transaction categorization for generating fiscal and budgetary reports.

Link: <https://github.com/AdrianLopez001/FINTRACK-SYSTEM>

LANGUAGES

- **Portuguese:** Native
- **English:** Intermediate (technical reading and documentation)
- **Spanish:** Basic

ADDITIONAL INFORMATION

- **Internship availability:** morning and afternoon
- **Areas of interest:** backend development, AI/Machine Learning, data, automation